A Comprehensive Analysis of AI-Based Detection of Data Poisoning Attacks in Machine Learning Systems

Vidya Reddy
School of Computer Science & Engineering
REVA University, Yelahanka, Bengaluru, India

Likitha
School of Computer Science & Engineering
REVA University, Yelahanka, Bengaluru, India

Sanchitha
School of Computer Science & Engineering
REVA University, Yelahanka, Bengaluru, India

Kshama
School of Computer Science & Engineering
REVA University, Yelahanka, Bengaluru, India

Varsha
School of Computer Science & Engineering
REVA University, Yelahanka, Bengaluru, India

***Abstract*—Machine learning systems are widely used in healthcare, finance, cybersecurity, autonomous systems, and recommendation applications, but their dependence on large training datasets creates important security risks. Data poisoning attacks intentionally insert, modify, or manipulate training samples to influence model behaviour. Such attacks can reduce accuracy, cause targeted misclassification, and introduce hidden backdoors. This paper presents a comprehensive analysis of AI-based approaches for detecting and mitigating data poisoning according to attack type, learning environment, detection technique, and defense mechanism. The reviewed approaches include anomaly detection, clustering, data-provenance analysis, weighted-average analysis, robust training, continuous monitoring, and federated learning. Representative results on CIFAR-10, Insurance Claims, Weka Weather, and Glass Identification datasets are compared, with emphasis on accuracy, precision, recall, attack success, and modification detection. The analysis also discusses challenges related to scalability, computational cost, stealthy poisoned samples, and adaptation to evolving attacks.**

***Index Terms*—data poisoning, machine learning security, artificial intelligence, anomaly detection, deep learning, backdoor attacks, data provenance, robust training, federated learning.**

1. Introduction

Machine learning (ML) has become an important technology in healthcare, finance, cybersecurity, autonomous systems, and recommendation systems because of its ability to learn complex patterns from large datasets. The increasing availability of large-scale data has enabled ML models to provide accurate predictions in many applications. However, dependence on training data also creates security vulnerabilities when data is collected from untrusted or multiple sources. Maintaining training-data integrity is therefore essential for reliable ML systems.

Data poisoning attacks are a significant threat in which an attacker intentionally inserts, modifies, or manipulates training samples to influence the learning process. Poisoned samples may reduce overall model performance, cause targeted misclassification, or introduce hidden backdoors. Common categories include availability attacks, targeted attacks, label-flipping attacks, clean-label attacks, feature poisoning, and backdoor attacks. Prior work has demonstrated that even a small number of manipulated samples can substantially influence a learned model [25], [26], [30].

Backdoor attacks are particularly difficult to identify because a poisoned model may behave normally on clean inputs but produce incorrect predictions when a specific trigger is present. Traditional data-cleaning techniques may also fail when malicious samples are designed to closely resemble legitimate training data. Consequently, AI-based detection techniques have gained importance for identifying abnormal patterns in data and learned model representations.

The remainder of this study is structured as follows. Section 2 presents related works and summarizes important attack and defense strategies. Section 3 describes the taxonomy of data poisoning attacks, datasets, and AI-based detection approaches. Section 4 presents the comparative analysis of representative methods and discusses their reported improvements. Section 5 presents the problem statement and the major challenges identified from the reviewed literature. Section 6 summarizes the findings and concludes the study with future research directions.

1.1. Background and Motivation

Backdoor attacks can preserve normal model accuracy on clean inputs while causing targeted errors when an attacker-controlled trigger appears. This property makes them difficult to detect using only conventional validation accuracy. Research on dataset security, poisoning, and backdoor attacks therefore emphasizes the need to inspect both training data and model behaviour [7], [8], [17].

1.2. Types of Data Poisoning Attacks

Poisoning attacks may target model availability, selected classes, labels, input features, or hidden model behaviour. Label-flipping attacks alter class labels, targeted attacks attempt to influence selected predictions, backdoor attacks insert a trigger that causes a specific incorrect output, and feature poisoning changes input attributes while attempting to remain difficult to distinguish from normal samples. Clean-label and targeted poisoning have further demonstrated that attacks can remain effective while preserving apparently valid labels [23], [26].

1.3. Detection Challenge

The main detection challenge is distinguishing malicious samples from legitimate data without removing useful training information. Detection approaches therefore examine statistical properties, feature-space distributions, learned representations, data provenance, and model behaviour. Techniques discussed in the reviewed studies include anomaly detection, clustering, influence analysis, spectral analysis, weighted-average analysis, robust training, and continuous monitoring [2], [7], [17].

2. Related Work

Recent research has investigated attack strategies, detection techniques, and defense mechanisms for poisoned training data. The studies reviewed in this paper are grouped according to attack and detection strategies, defense mechanisms, AI-based detection, and application-specific research.

2.1. Attack Taxonomy and Detection Strategies

Biggio, Nelson, and Laskov demonstrated that manipulated training samples can influence the decision boundary of support vector machines and degrade model performance [30]. Subsequent work extended poisoning to deep learning and demonstrated back-gradient optimization for constructing effective poisoning samples [25]. Targeted backdoor poisoning showed that a small number of carefully constructed samples can implant hidden behaviour into a deep model [26]. BadNets further illustrated the risk of outsourced or third-party model training [27].

2.2. Defense Mechanisms

Robust data validation and continuous monitoring can identify suspicious changes before they significantly affect a model. Data provenance records the origin and transformation history of training data, improving traceability. Certified defenses and robust optimization provide formal or empirical protection against bounded poisoning budgets [20], [24], [28]. Multi-layered approaches combine sanitization, robust training, cryptographic verification, and monitoring to protect the training lifecycle [5].

2.3. AI-Based Detection Approaches

The Verified Weighted Average (VWA) approach evaluates weighted averages of input features and compares calculated values with expected values. The reviewed study evaluated VWA using the Weka Weather and Glass Identification datasets and compared it with other detection strategies [2]. Neural Cleanse and STRIP provide complementary examples of backdoor detection based on trigger reconstruction and runtime entropy behaviour [17], [18].

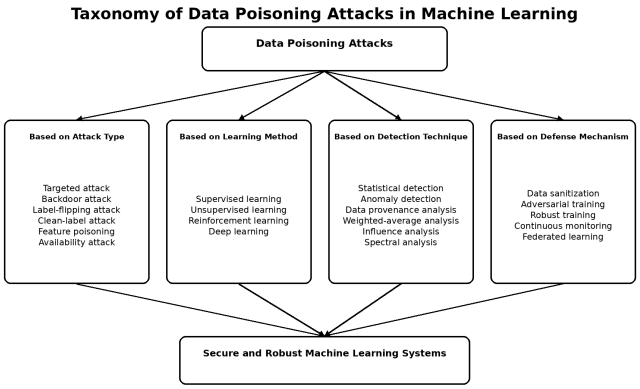


Fig. 1. Taxonomy of data poisoning attacks and detection approaches in machine learning.

2.4. Domain-Specific Studies

Data poisoning has also been studied in intelligent transportation, malware detection, autoregressive models, and distributed learning environments. Intelligent transportation systems can be affected through multiple vulnerable data sources [3]. Targeted poisoning has also been demonstrated against malware detectors [13], while federated learning introduces additional poisoning surfaces because participating clients influence distributed training [15], [16].

3. Taxonomy, Datasets, and Detection Approaches

Figure 1 summarizes the reviewed taxonomy according to attack type, learning method, detection technique, and defense mechanism. The categories lead toward the common objective of secure and robust machine learning systems.

3.1. Datasets

The reviewed studies use different datasets to evaluate poisoning effects and detection. CIFAR-10 is used for image-classification experiments, Insurance Claims for fraud detection, Weka Weather for binary classification and weighted-average analysis, and Glass Identification for evaluating possible data modification. Figure 2 presents the reviewed dataset landscape.

3.2. Insurance Claims Dataset

The Insurance Claims dataset is used to investigate poisoning in a fraud-detection application. The reviewed experiment includes data cleaning, categorical encoding, poisoning simulation, training/test separation, Random Forest training, and comparison using accuracy, classification reports, and confusion matrices. Figure 3 shows the evaluation workflow.

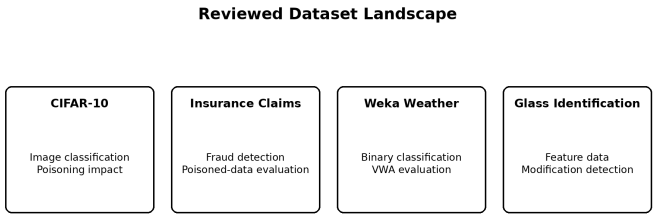


Fig. 2. Datasets discussed in the reviewed data-poisoning studies.

TABLE 1
SUMMARY OF DATASETS USED IN THE REVIEWED STUDIES

Dataset	Application	Purpose	Evaluation
CIFAR-10	Image classification	Poisoning impact	Accuracy comparison
Insurance Claims	Fraud detection	Poisoned-data evaluation	Accuracy, reports, confusion matrix
Weka Weather	Binary classification	VWA evaluation	Weighted-average comparison
Glass Identification	Feature classification	Modification evaluation	λ-based detection

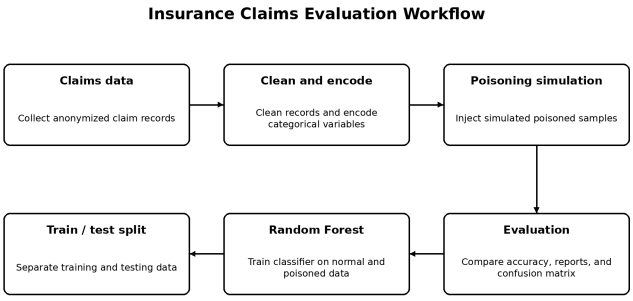


Fig. 3. Redrawn Insurance Claims evaluation workflow.

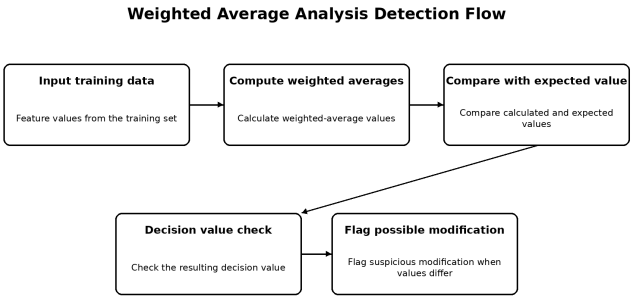


Fig. 4. Redrawn weighted-average analysis detection flow.

3.3. Weighted Average Analysis Detection

The VWA method calculates weighted averages of input features and compares the resulting value with an expected value. A deviation can indicate possible manipulation. The approach supports binary and multiclass classification and was evaluated using Weka Weather and Glass Identification datasets [2]. Figure 4 summarizes the detection flow.

3.4. Hybrid and Layered Defense

Hybrid models combine two or more detection or defense techniques to improve the identification and mitigation of data poisoning. Data sanitization, robust training, continuous monitoring, provenance tracking, and distributed protection provide complementary safeguards. Figure 5 presents the layered defense concept.

4. Comparative Analysis

The existing studies use different approaches to identify and mitigate data poisoning attacks. One group evaluates the impact of poisoning using anomaly detection, robust optimization, adversarial training, and ensemble learning, whereas another group focuses on direct weighted-average analysis for detecting possible modifications. Detection methods differ in their detection stage, computational requirements, and type of evidence used to identify suspicious data.

4.1. Comparison of Two Representative Approaches

Table II compares two representative approaches selected from the reviewed material. The first demonstrates how poisoning can degrade model performance and how defense mechanisms can improve robustness. The second focuses on detecting possible modification through a direct weighted-average test.

4.2. Key Findings

Three broad findings emerge. First, poisoning can cause large accuracy losses, so clean-data performance alone is not sufficient. Second, detection effectiveness depends on dataset characteristics and attack type. Third, layered defenses combining sanitization, robust training, monitoring, and distributed protection provide complementary safeguards across the ML lifecycle [7], [10], [12], [13].

5. Problem Statement

The reviewed literature identifies three focused problems:

- **Reliable identification:** Poisoned samples can closely resemble legitimate training data, making malicious modifications difficult to distinguish without reducing useful data.
- **Scalable and adaptable detection:** Existing methods vary in effectiveness across datasets and attack types, while large and complex datasets increase computational demands.
- **Balanced protection:** A practical framework must detect and reduce poisoning while maintaining model accuracy, reliability, and scalability against evolving attacks.

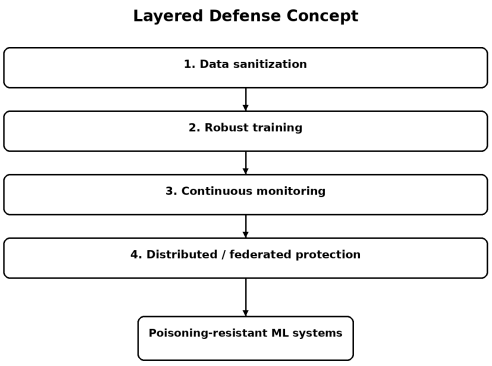


Fig. 5. Redrawn layered defense concept for poisoning-resistant machine learning.

TABLE 2
COMPARATIVE ANALYSIS OF TWO REPRESENTATIVE DATA-POISONING APPROACHES

Aspect	Poisoning-effect study	VWA approach
Goal	Measure poisoning impact and evaluate detection/defense	Detect possible training-data modification
Methods	Anomaly detection, robust optimization, adversarial training, ensemble learning	Verified Weighted Average analysis
Datasets	CIFAR-10; Insurance Claims	Weka Weather; Glass Identification
Key result	CIFAR-10: 92.3% to 65.1%; Insurance Claims: 97.2% to 74.5%	Reported accuracy 0.92, precision 0.94, recall 0.91; modification detected in Glass
Contribution	Demonstrates poisoning impact and evaluates robustness	Provides a direct weighted-average modification check
Limitation	Effectiveness depends on attack and dataset	Performance may vary with dataset and attack type

6. Conclusion

Data poisoning attacks pose a significant threat to the reliability and security of ML systems because manipulated training data can affect model learning and lead to incorrect predictions. The reviewed studies cover CIFAR-10, Insurance Claims, Weka Weather, and Glass Identification and examine anomaly detection, statistical analysis, data sanitization, robust training, ensemble learning, data provenance, and VWA. Reported results show that poisoning can considerably reduce model performance; representative studies report substantial degradation on image-classification and fraud-detection tasks. Overall, effective protection requires reliable validation, continuous detection, robust training, and adaptable defenses. Future work can focus on scalable defenses, federated learning, explainable AI, privacy-preserving techniques, and evaluation on larger and more diverse datasets.

Acknowledgment

The authors acknowledge the researchers and publications reviewed in this study for providing the methods, datasets, and experimental findings discussed in the taxonomy, comparative analysis, and defense mechanisms.

References

[1] B. Nazer, P. Savadattimath, and S. Kumari, "Data poisoning attacks and defenses in machine learning: A survey with focus on computer vision," in *Proc. IEEE Int. Conf. Computer Vision and Machine Intelligence (CVMI)*, 2025, pp. 1–6.

[2] Y. R. Maramreddy and K. Muppavaram, "Detecting and mitigating data poisoning attacks in machine learning: A weighted average approach," *Eng., Technol. Appl. Sci. Res.*, vol. 14, no. 4, pp. 15505–15509, 2024.

[3] F. Wang, X. Wang, and X. J. Ban, "Data poisoning attacks in intelligent transportation systems: A survey," *Transp. Res. C*, vol. 165, Art. no. 104750, 2024.

[4] T. Ino, K. Yoshida, H. Matsutani, and T. Fujino, "Data poisoning attack against neural network-based on-device learning anomaly detector by physical attacks on sensors," *Sensors*, vol. 24, no. 19, Art. no. 6416, 2024.

[5] G. Keshavarzkalhori, C. Perez-Sola, G. Navarro-Arribas, and J. Herrera-Joancomarti, "Building resilient AI: A solution to data and model poisoning prevention," in *Proc. 17th Int. Conf. Security of Information and Networks (SIN)*, 2024, pp. 1–8.

[6] S. Sardana, S. Gupta, A. Donode, A. Prasad, and G. M. Karthik, "Defending machine learning and deep learning models: Detecting and preventing data poisoning attacks," in *Proc. Global Conf. Communications and Information Technologies (GCCIT)*, 2024, pp. 100–105.

[7] M. Goldblum, D. Tsipras, C. Xie, X. Chen, A. Schwarzschild, D. Song, A. Madry, B. Li, and T. Goldstein, "Dataset security for machine learning: Data poisoning, backdoor attacks, and defenses," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 1563–1580, 2023.

[8] Z. Wang, J. Ma, X. Wang, J. Hu, Z. Qin, and K. Ren, "Threats to training: A survey of poisoning attacks and defenses on machine learning systems," *ACM Comput. Surveys*, vol. 55, no. 7, Art. no. 134, pp. 1–36, 2023.

[9] Y. Gao, Y. Kim, B. G. Doan, Z. Zhang, G. Zhang, S. Nepal, D. C. Ranasinghe, and H. Kim, "Design and evaluation of a multi-domain Trojan detection method on deep neural networks," *IEEE Trans. Dependable Secure Comput.*, vol. 19, pp. 2349–2364, 2022.

[10] C. Wang, J. Chen, Y. Yang, X. Ma, and J. Liu, "Poisoning attacks and countermeasures in intelligent networks: Status quo and prospects," *Digital Communications and Networks*, vol. 8, no. 2, pp. 225–234, 2022.

[11] Z. Zhang, A. Panda, L. Song, Y. Yang, M. Mahoney, J. Gonzalez, K. Ramchandran, and P. Mittal, "Neurotoxin: Durable backdoors in federated learning," in *Proc. Int. Conf. Machine Learning*, 2022, pp. 26429–26446.

[12] D. Wu and Y. Wang, "Adversarial neuron pruning purifies backdoored deep models," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 16913–16925.

[13] S. Narisada, S. Sasaki, S. Hidano, T. Uchibayashi, T. Suganuma, M. Hiji, and S. Kiyomoto, "Stronger targeted poisoning attacks against malware detection," in *Proc. Cryptology and Network Security*, 2020, pp. 65–84.

[14] B. G. Doan, E. Abbasnejad, and D. C. Ranasinghe, "Februus: Input purification defense against Trojan attacks on deep neural network systems," in *Proc. 36th Annual Computer Security Applications Conf. (ACSAC)*, 2020, pp. 897–912.

[15] M. Fang, X. Cao, J. Jia, and N. Z. Gong, "Local model poisoning attacks to Byzantine-robust federated learning," in *Proc. 29th USENIX Security Symp.*, 2020, pp. 1623–1640.

[16] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How to backdoor federated learning," in *Proc. Int. Conf. Artificial Intelligence and Statistics*, 2020, pp. 2938–2948.

[17] B. Wang, Y. Yao, S. Shan, H. Li, B. Viswanath, H. Zheng, and B. Y. Zhao, "Neural Cleanse: Identifying and mitigating backdoor attacks in neural networks," in *Proc. IEEE Symp. Security and Privacy*, 2019, pp. 707–723.

[18] Y. Gao, C. Xu, D. Wang, S. Chen, D. C. Ranasinghe, and S. Nepal, "STRIP: A defence against Trojan attacks on deep neural networks," in *Proc. 35th Annual Computer Security Applications Conf.*, 2019, pp. 113–125.

[19] A. N. Bhagoji, S. Chakraborty, P. Mittal, and S. Calo, "Analyzing federated learning through an adversarial lens," in *Proc. Int. Conf. Machine Learning*, 2019, pp. 634–643.

[20] M. Jagielski, A. Oprea, B. Biggio, C. Liu, C. Nita-Rotaru, and B. Li, "Manipulating machine learning: Poisoning attacks and countermeasures for regression learning," in *Proc. IEEE Symp. Security and Privacy*, 2018, pp. 19–36.

[21] K. Liu, B. Dolan-Gavitt, and S. Garg, "Fine-pruning: Defending against backdooring attacks on deep neural networks," in *Proc. Research in Attacks, Intrusions, and Defenses*, 2018, pp. 273–294.

[22] E. M. El-Mhamdi, R. Guerraoui, and S. Rouault, "The hidden vulnerability of distributed learning in Byzantium," in *Proc. Int. Conf. Machine Learning*, vol. 80, 2018, pp. 3521–3530.

[23] A. Shafahi, W. R. Huang, M. Najibi, O. Suciu, C. Studer, T. Dumitras, and T. Goldstein, "Poison frogs! Targeted clean-label poisoning attacks on neural networks," in *Advances in Neural Information Processing Systems*, vol. 31, 2018, pp. 6106–6116.

[24] J. Steinhardt, P. W. Koh, and P. S. Liang, "Certified defenses for data poisoning attacks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3520–3532.

[25] L. Muñoz-González, B. Biggio, A. Demontis, A. Paudice, V. Wongrassamee, E. C. Lupu, and F. Roli, "Towards poisoning of deep learning algorithms with back-gradient optimization," in *Proc. 10th ACM Workshop on Artificial Intelligence and Security*, 2017, pp. 27–38.

[26] X. Chen, C. Liu, B. Li, K. Lu, and D. Song, "Targeted backdoor attacks on deep learning systems using data poisoning," arXiv:1712.05526, 2017, pp. 1–18.

[27] T. Gu, B. Dolan-Gavitt, and S. Garg, "BadNets: Identifying vulnerabilities in the machine learning model supply chain," arXiv:1708.06733, 2017, pp. 1–13.

[28] N. Baracaldo, B. Chen, H. Ludwig, and J. A. Safavi, "Mitigating poisoning attacks on machine learning models: A data provenance based approach," in *Proc. ACM Workshop on Artificial Intelligence and Security*, 2017, pp. 103–110.

[29] M. Mozaffari-Kermani, S. Sur-Kolay, A. Raghunathan, and N. K. Jha, "Systematic poisoning attacks on and defenses for machine learning in healthcare," *IEEE J. Biomed. Health Inform.*, vol. 19, no. 6, pp. 1893–1905, 2015.

[30] B. Biggio, B. Nelson, and P. Laskov, "Poisoning attacks against support vector machines," in *Proc. 29th Int. Conf. Machine Learning*, 2012, pp. 1467–1474.